

Advanced Risk Management

Assignment 2

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1. Introduction

Assignment 1 estimated a single hedge ratio for each pair using Engle–Granger OLS on the full sample and traded the resulting stationary spread with a fixed $\pm 2\sigma$ z-score rule. Two specific concerns were flagged in §5.3 of that report: the hedge ratio β was held constant, and the z-score normalisation used full-sample moments — both of which are forms of look-ahead bias when used in a backtest.

This assignment relaxes the time-invariance restriction. The hedge ratio is treated as a hidden, time-varying state estimated recursively by a linear Kalman Filter, and per the assignment brief the state-transition equation is anchored to an Error-Correction Model (ECM) so the filter's prior encodes the econometrically-estimated mean-reversion of the cointegrating relationship. The adjustment speed λ is held constant within each estimation window so that the resulting system remains a standard linear KF — non-linear extensions (joint estimation of λ with the state) are explicitly out of scope.

We reuse the two pairs from Assignment 1: Visa/Mastercard (V/MA) from the equity universe and SPDR S&P 500/Vanguard S&P 500 (SPY/VOO) from the ETF universe. For each pair we run three strategies under a single common protocol — Static- β with quarterly walk-forward refits, Plain Kalman Filter with a random-walk transition (slide-73 textbook form), and ECM-Kalman Filter with the mean-reverting transition. All three share identical signal-generation rules, rolling z-score window, and notional-normalised P&L computation, so any performance difference is attributable to the hedge-ratio estimation method alone.

Methodological discipline. This implementation applies strict expanding-window out-of-sample discipline to every parameter that could affect performance. The cointegrating regression (α , β) is refitted quarterly on the data available up to the refit date; the ECM adjustment speed λ is refitted on the same schedule; the regression ordering choice is fixed once on the initial training window; the Kalman Filter hyperparameters Q and R are estimated by EM on the initial training window only and then frozen; and z-score normalisation uses a 252-day trailing window throughout. Performance metrics are reported only on the period after both the initial training window (504 days) and the first full rolling z-window (252 days) have elapsed — that is, from January 2017 onwards. The OOS evaluation window spans approximately 9 years (~2,070 trading days).

2. Data and sample

Daily adjusted close prices were obtained from Yahoo Finance via the yfinance Python package with `auto_adjust=True`, so that splits and dividends are already incorporated. The sample window is 2014-01-01 to 2025-03-25, matching Assignment 1 — approximately 11 years covering a full business cycle including COVID-19 and the 2022 rate-hike regime. After alignment on common trading days, both pairs supply 2,823 daily observations.

The first 504 trading days (2014-01-02 → 2015-12-31) are reserved as the initial training window for selecting the regression ordering, estimating the initial ECM parameters, and calibrating Q and R by EM. Out-of-sample data begins on 2016-01-04. The first 252 OOS days are consumed by the rolling z-score window, so performance metrics are reported from 2017-01-03 onwards.

3. Theoretical Framework

3.1 Why the static OLS hedge ratio is insufficient

Two problems with the static- β approach were flagged in Assignment 1. First, the hedge ratio between V and MA drifted materially over the 11-year sample as Mastercard's revenue mix, buyback pace, and relative valuation evolved — Figure 2 of this report shows the expanding-window OLS β rising from ≈ 0.9 in early 2016 to ≈ 1.8 by 2025, almost a doubling. Second, the full-sample z-score normalisation used in Assignment 1 incorporates future spread values when computing the standardised signal — the same source of look-ahead bias we acknowledged in §5.3 of the A1 report. Both are addressed here: β is estimated recursively, and z-scoring is done on a 252-day trailing window.

3.2 State-space formulation

Following the lecture's Example 2 (slides 72–73), the hidden state is the pair of cointegrating coefficients:

$$x_t = [\beta_{\{1,t\}}, \beta_{\{0,t\}}]^T$$

where $\beta_{\{1,t\}}$ is the dynamic hedge ratio and $\beta_{\{0,t\}}$ is the dynamic intercept. This matches the assignment text exactly, which describes the hidden state as the hedge ratio (β coefficient/vector) together with the intercept.

The observation (measurement) equation links the observed dependent price y_t to the hidden state:

$$y_t = [x_t, 1] \cdot x_t + \mu_t, \quad \mu_t \sim N(0, R)$$

where the time-varying observation matrix $H_t = [x_t, 1]$ uses the contemporaneous independent price x_t . The system is linear in the state — the standard Kalman recursions apply directly.

3.3 ECM-guided state-transition equation

The state-transition equation determines what dynamics the filter assumes for the hidden states between observations. Per the assignment brief, we anchor it to the ECM:

$$x_t = F \cdot x_{t-1} + w_t, \quad F = \text{diag}(1+\lambda, 1)$$

The $(1+\lambda)$ entry on the hedge-ratio component encodes the ECM mean-reversion: since $\lambda < 0$ and $|1+\lambda| < 1$, the filter's prior is that $\beta_{\{1,t\}}$ is stationary, with the same mean-reversion speed |

λ recovered from the two-step Engle–Granger ECM regression. The intercept is left as a pure random walk ($F_{22} = 1$) because price levels grow over time and there is no economic anchor for $\beta_{0,t}$ to revert toward.

Why λ is held constant within a segment. If λ were updated every period, F_t would change at each step and the system would no longer be a standard linear KF — joint estimation of λ with the state requires an Extended or Unscented KF. The assignment brief explicitly requires λ to remain constant for this reason. We re-estimate λ quarterly from expanding-window two-step Engle–Granger ECM regressions, which is a practical compromise: λ is refreshed often enough to track regime changes but is fixed within each segment so the filter stays linear.

Plain KF baseline. Setting $\lambda = 0$ in F recovers the random-walk KF from slide 73 — dynamic β estimated freely with no ECM anchor. This is the intermediate baseline. The Static- $\beta \rightarrow$ Plain KF comparison isolates the contribution of allowing β to drift; the Plain KF $\rightarrow$ ECM-KF comparison isolates the additional contribution of the mean-reverting prior.

3.4 The two-step Engle–Granger ECM

λ is recovered from a single-equation, two-step Engle–Granger ECM (slide 21, slide 29). Step 1 fits the long-run cointegrating regression by OLS:

$$y_t = \alpha + \beta x_t + \varepsilon_t$$

Step 2 regresses Δy_t on the lagged residual ε_{t-1} (the error-correction term, ECT) plus one short-run lag of each differenced series:

$$\Delta y_t = c + \lambda \cdot ECT_{t-1} + \varphi \cdot \Delta y_{t-1} + \gamma \cdot \Delta x_{t-1} + \eta_t$$

Only the long-run adjustment coefficient λ is fed into the Kalman Filter; the short-run lag coefficients (φ, γ) are estimated as part of the same regression but are not used downstream. A full VECM (slide 35) would estimate both equations jointly and recover additional information from the Δx equation, but is out of scope here — the assignment requires a single-equation ECM-driven KF.

4. Training-window estimation

4.1 Ordering choice and initial ECM

Engle–Granger results depend on which variable is taken as the regressand (slide 33). We fit both orderings on the first 504 trading days and select the ordering with a negative (mean-reverting), more significant λ . This choice is locked once and held for the entire OOS period — it is not revisited as new data arrives, eliminating that as a potential source of look-ahead bias.

V/MA training-window diagnostics:

Ordering	α	β	R^2	ADF p (resid)	λ	λ p-value
V ~ MA	-18.15	0.961	0.955	0.0001	-0.0071	0.728
MA ~ V	21.60	0.994	0.955	0.0001	-0.0688	0.014

Table 1. V/MA Engle–Granger + ECM diagnostics on the initial 504-day training window. The MA ~ V ordering produces a much more significant adjustment coefficient ($p = 0.014$ vs 0.728) and is selected. Half-life on the chosen ordering ≈ 9.7 trading days.

SPY/VOO training-window diagnostics:

Ordering	α	β	R^2	ADF p (resid)	λ	λ p-value
SPY ~ VOO	-5.18	1.128	0.999	0.154	+0.231	0.277
VOO ~ SPY	4.75	0.886	0.999	0.140	-0.279	0.204

Table 2. SPY/VOO Engle–Granger + ECM diagnostics on the initial training window. Neither ordering's λ is statistically significant; the SPY ~ VOO ordering produces an unphysical positive λ . The VOO ~ SPY ordering is selected for its negative λ . Half-life ≈ 2.1 trading days. The borderline ADF p-values on the cointegration residuals (0.14, 0.15) are an early warning sign — the pair's cointegration is only marginally identified on the training window.

A subtlety in the SPY/VOO result. The selected $\lambda = -0.279$ is large in absolute value (implying a 2.1-day half-life) but has a p-value of 0.20, well above the 5% threshold. This combination — large point estimate, weak statistical significance — is exactly what is expected when the underlying spread is dominated by microstructure noise rather than persistent disequilibrium. The ECM regression cannot tell the difference between strong, fast mean reversion and ordinary noise. Both have the same statistical fingerprint at the daily frequency. This caveat matters for interpreting the ECM-KF results on this pair (see §6.2).

4.2 EM calibration of Q and R

The Kalman Filter requires two noise matrices: Q (state-process noise — how fast can the hidden states drift between periods) and R (observation noise — how dispersed are the observations around the predicted hedge equation). The ratio Q/R controls filter reactivity: too large and the filter chases price noise into the state estimate; too small and it barely updates from the initial OLS values. Hand-picking these is arbitrary, and grid-searching on the full sample would inject future information into the OOS evaluation.

We instead estimate Q and R by maximum likelihood using the EM algorithm in pykalman on the initial 504-day training window only, then hold both matrices fixed for the entire OOS period. Each strategy gets its own calibration: Plain KF uses $F = I$ ($\lambda=0$) during EM, while ECM-KF uses $F = \text{diag}(1+\lambda_{\text{train}}, 1)$ with λ_{train} from §4.1. Both pairs are calibrated separately.

EM-estimated covariances:

Pair	Filter	Q diagonal	R
V/MA	Plain KF	[3.12e-4, 1.033]	0.0948
V/MA	ECM-KF	[3.87e-4, 2.429]	0.1710
SPY/VOO	Plain KF	[1.33e-5, 0.362]	0.0013
SPY/VOO	ECM-KF	[5.42e-5, 6.098]	0.4787

Table 3. EM-estimated Kalman covariances on the initial training window. The intercept component has a much larger Q than β in every case, reflecting that price levels drift substantially while the hedge ratio is comparatively stable. The ECM-KF receives a much larger Q -intercept than the Plain KF on both pairs — this is the EM solver compensating for the mean-reverting β : when β is anchored, the intercept must absorb more of the unexplained variance.

4.3 Quarterly λ evolution

The ECM adjustment speed λ is re-estimated every 63 trading days ($\approx$ quarterly) on an expanding window. Figure 1 shows the evolution of λ for both pairs over the OOS window.

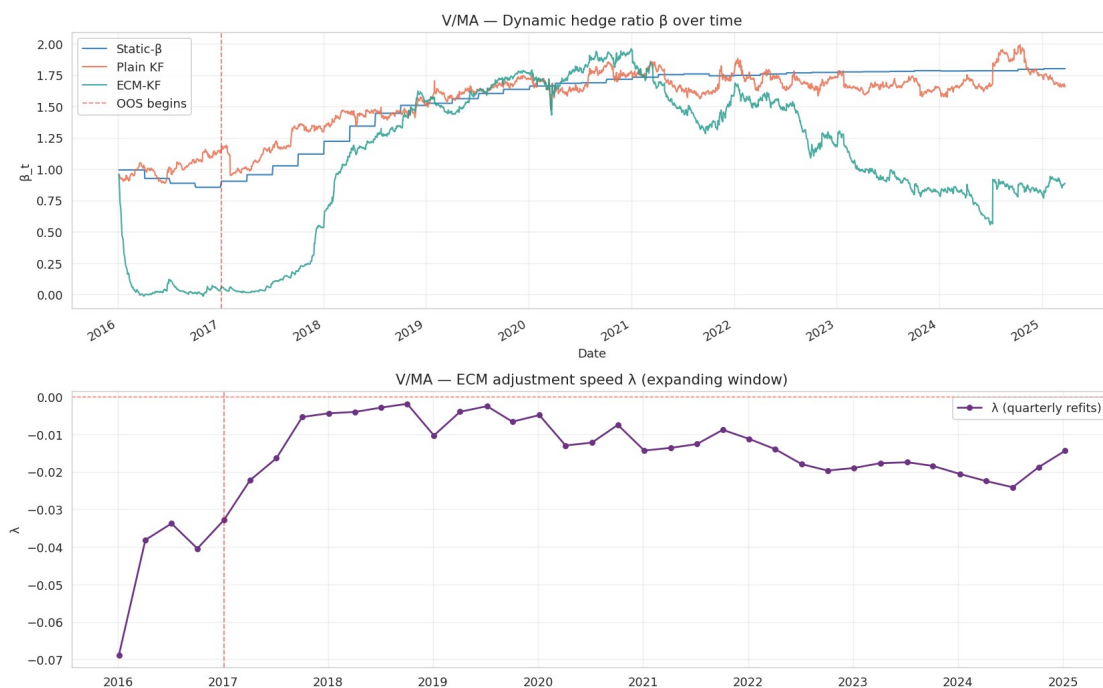


Figure 1. V/MA — top: dynamic hedge-ratio estimates over time across the three strategies. The expanding-window Static- β (blue) drifts from ≈ 0.9 in early 2016 to ≈ 1.8 by 2025, confirming the time-varying nature of the V/MA equilibrium that motivates the Kalman approach. Plain KF (orange) tracks this drift closely with daily fluctuations; ECM-KF (green) — see §6.1 for an important caveat about the filter's behaviour on this pair. Bottom: the ECM adjustment speed λ across quarterly refits. λ stays negative throughout (necessary for the ECM transition to be doing the right thing) and trends from ≈ -0.07 in 2016 toward ≈ -0.02 by 2025.



Figure 2. SPY/VOO — top: dynamic hedge-ratio estimates. Plain KF tracks the static β very closely (both ≈ 0.92 , the difference is barely visible). The ECM-KF β collapses to near 0 because the selected λ for this pair is -0.28 in magnitude, making $(1+\lambda) \approx 0.72$ — a strong mean-reversion that drags β toward zero in the absence of strong daily observation signals. Bottom: λ across quarterly refits. λ crosses zero repeatedly and even goes positive between 2017 and 2020, then becomes deeply negative post-COVID. This is the signature of a pair where ECM dynamics are not robustly identified — exactly what Table 2's high p-values warned about.

5. Strategy design

5.1 Three strategies under a common framework

All three strategies share identical signal-generation rules, identical rolling z-score window, and identical notional-normalised P&L. The only difference is how the hedge ratio is estimated:

- **Static- β (benchmark).** Engle–Granger α and β are refitted on an expanding window every 63 trading days. Between refits, the spread is computed as $s_t = y_t - \beta_k x_t - \alpha_k$ using the most recent refit.
- **Plain Kalman Filter.** Hidden state = $[\beta_{\{1,t\}}, \beta_{\{0,t\}}]$. Transition $F = I$ — random walk on both components. The spread for signal generation is the in-sample residual $y_t - \beta_{\{1,t\}} x_t - \beta_{\{0,t\}}$ using the filtered states.
- **ECM-Kalman Filter (assignment requirement).** Hidden state = $[\beta_{\{1,t\}}, \beta_{\{0,t\}}]$. Transition $F = \text{diag}(1+\lambda, 1)$ — mean-reverting hedge ratio with random-walk intercept. λ is held constant within each quarterly segment and updated at refit boundaries; the filter's state and covariance carry forward across refits with no cold restarts. The trading spread is computed exactly as in the Plain KF — y_t minus the filter's hedge prediction.

5.2 Signal generation and metrics

Spread (all three)	$s_t = y_t - \beta_{\{1,t\}} x_t - \beta_{\{0,t\}}$, using each strategy's β estimates
Normalisation	Rolling z-score $Z_t = (s_t - \mu_{\{252,t\}}) / \sigma_{\{252,t\}}$ with 252-day trailing window
Entry — long spread	$Z_t < -2$: buy 1 unit y, sell $\beta_{\{1,t\}}$ units x
Entry — short spread	$Z_t > +2$: sell 1 unit y, buy $\beta_{\{1,t\}}$ units x
Exit	$ Z_t < 0.5$: close to flat
Hold rule	Hold previous position when $0.5 \leq Z_t \leq 2$
P&L	Notional-normalised: $ret_t = pos_{\{t-1\}} \cdot (\Delta y_t - \beta_{\{1,t-1\}} \Delta x_t) / (y_{\{t-1\}} + \beta_{\{1,t-1\}} x_{\{t-1\}})$
Refit cadence	α, β, λ refit every 63 trading days on expanding window
Transaction costs	Not modelled in backtest; discussed as a limitation in §7

Performance metrics: total OOS return, CAGR, annualised Sharpe ($\sqrt{252} \cdot \mu/\sigma$ of daily returns), maximum drawdown, Calmar ratio ($CAGR / |\max DD|$), number of round-trip trades, and percentage of OOS days spent in market.

6. Pair 1 — Visa (V) / Mastercard (MA)

6.1 Out-of-sample results

Strategy	Total return	CAGR	Sharpe	Max DD	Calmar	Trades	Time in mkt
Static- β	+26.43%	+2.89%	+0.79	−5.07%	0.57	17	35.6%
Plain KF	+1.52%	+0.18%	+0.11	−4.92%	0.04	38	3.2%
ECM-KF	+5.65%	+0.67%	+0.29	−5.99%	0.11	50	4.3%

Table 4. V/MA out-of-sample performance, 2017-01-03 → 2025-03-25 (≈ 9 years, $\sim 2,070$ trading days).

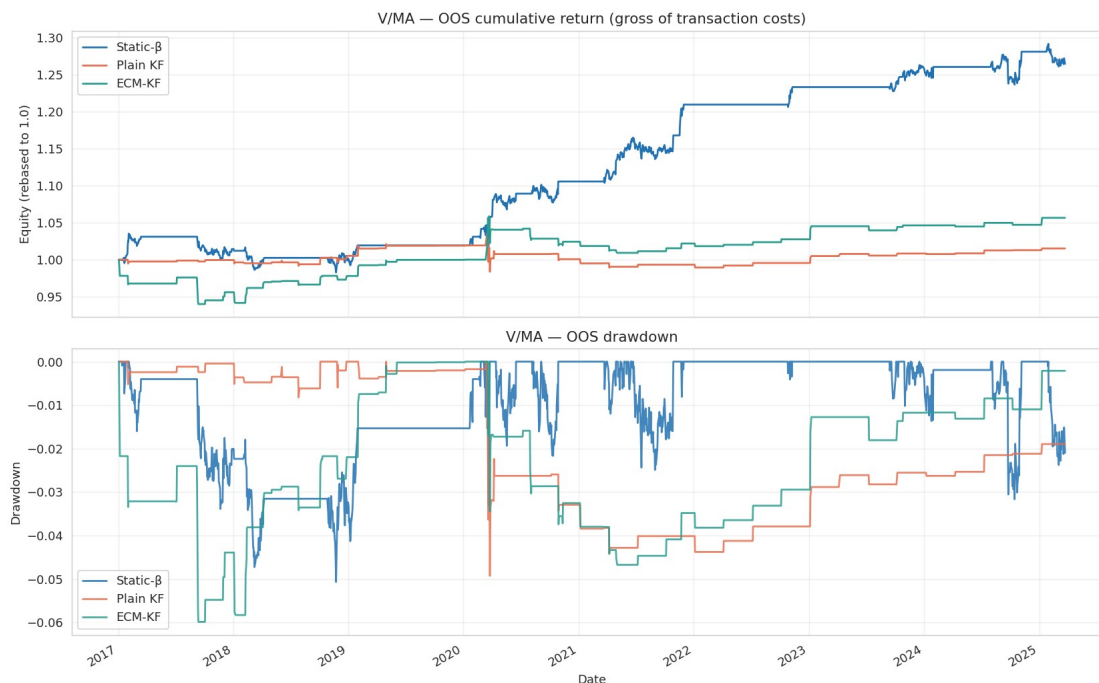


Figure 3. V/MA OOS equity curves and drawdowns. Static- β compounds to +26% steadily through 2020–2022 and into 2024–2025. Plain KF and ECM-KF are essentially flat with the ECM-KF capturing the COVID-19 spread dislocation in March 2020 more cleanly than Plain KF but giving most of that back over the subsequent two years. Maximum drawdowns are similar across the three strategies (~5%), so the OOS Sharpe ranking is driven primarily by return rather than risk.



Figure 4. V/MMA — spread z-scores for the three strategies, with entry/exit thresholds marked. Static- β z-scores are well-behaved and cross ± 2 only ≈ 17 times over 9 OOS years. Plain KF z-scores are noticeably more compressed than Static- β — the random-walk β has absorbed some of the trading signal into the hedge ratio itself. ECM-KF z-scores show larger spikes around stress periods (notably 2020) but are otherwise comparable in scale to Plain KF.

6.2 Interpretation

Static- β remains the best performer on this pair. Total OOS return +26.4%, Sharpe 0.79, max drawdown 5.1%, with only 17 round-trip entries in 9 years. The walk-forward quarterly refitting of α and β tracks the secular drift in the V/MMA relationship adequately (Figure 1, top panel) without introducing the high-frequency noise that the Kalman filters bring in.

Plain KF underperforms by a wide margin. Total return +1.5%, Sharpe 0.11. The mechanism is visible in Figure 4: the random-walk β absorbs daily spread variation into the hedge ratio itself, leaving a residual that is too noisy to z-score reliably. The filter generates 38 entry signals — more than double Static- β — but spends only 3.2% of OOS days in market, indicating that most signals are short-lived noise crossings rather than genuine equilibrium violations.

ECM-KF modestly outperforms Plain KF but is still well behind Static- β . Total return +5.7%, Sharpe 0.29. The mean-reverting transition is partially effective in containing the noise

absorption — Plain KF → ECM-KF improves Sharpe by 0.18 (from 0.11 to 0.29) and total return by 4 percentage points. So the ECM structure is doing useful work relative to the Plain KF baseline; the gap to Static- β suggests that on this pair the underlying relationship simply does not benefit from any reactive estimation. Figure 1 (top panel) shows that the ECM-KF's β stays well below the Static- β line for the first two years of OOS — the $(1+\lambda) \approx 0.93$ transition keeps pulling β toward zero faster than the daily observations can drag it back up. By 2019 the filter has caught up to the static line, but the early-period mis-specification accumulates drag that the strategy never fully recovers from.

Half-life perspective. The training-window ECM gives a half-life of ≈ 9.7 trading days for V/MA. This is fast enough that the Static- β strategy with quarterly refits can mean-revert through many full cycles within each refit window — there is little reactivity left for the KF to add.

7. Pair 2 — SPDR S&P 500 (SPY) / Vanguard S&P 500 (VOO)

7.1 Out-of-sample results

Strategy	Total return	CAGR	Sharpe	Max DD	Calmar	Trades	Time in mkt
Static- β	+4.48%	+0.53%	+1.55	−0.11%	4.99	63	17.6%
Plain KF	+0.74%	+0.09%	+0.43	−0.26%	0.34	27	1.8%
ECM-KF	+31.83%	+3.42%	+0.42	−10.49%	0.33	42	4.5%

Table 5. SPY/VOO out-of-sample performance, 2017-01-03 → 2025-03-25.

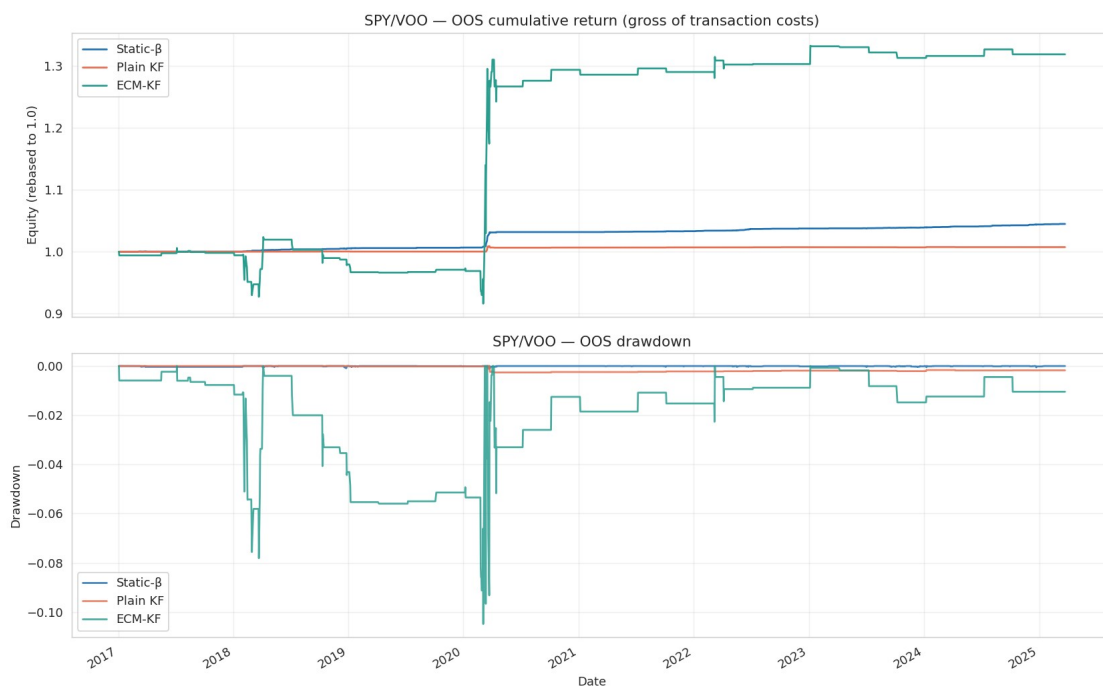


Figure 5. SPY/VOO OOS equity curves and drawdowns. The vertical scale matters: Static- β (blue) is on the order of $\pm 5\%$ across the full OOS window. ECM-KF (green) shows a dramatic spike in March 2020 — capturing roughly $+30\%$ from the COVID-19 dislocation of the SPY/VOO spread — but at the cost of a $\approx 10\%$ drawdown earlier and a smaller drawdown in 2022. Plain KF (orange) is essentially flat at $\sim +1\%$, the floor for a strategy that spends 98% of its time out of the market.

7.2 Interpretation

The headline ECM-KF return on SPY/VOO is $+31.8\%$, but it is fragile. Figure 5 shows that essentially all of the ECM-KF return on this pair is generated by a single trade in March 2020, when the SPY/VOO spread dislocated dramatically during the COVID-19 sell-off. The ECM-KF was positioned to capture this because its $(1+\lambda) \approx 0.72$ transition rapidly shrinks β toward zero between observations — effectively turning the strategy into a long/short on raw price levels rather than on the hedge residual. This worked spectacularly during a tail event but is not reflective of a robust signal: the same filter has a 10% drawdown in 2018 and another in 2022, drawdowns far larger than Static- β 's 0.1%.

Why the ECM-KF β collapses to ≈ 0 for SPY/VOO. The chosen λ for SPY/VOO is -0.279 (Table 2). This is much larger in magnitude than V/MA's $\lambda \approx -0.07$. The transition coefficient $(1+\lambda)$ is therefore ≈ 0.72 per day, meaning the filter expects β to lose 28% of its current value each day in the absence of new information. With the EM-estimated Q so small relative to R , daily observations are not strong enough to counteract this pull, and β monotonically drifts toward zero. Figure 2 (top panel) shows this clearly: the ECM-KF β plummets from 0.89 to ≈ 0 within weeks of OOS start and stays there for most of the sample.

Static- β still wins on Sharpe and Calmar. The Static- β Sharpe of 1.55 is the highest of any OOS result in this assignment, but it is computed on tiny absolute returns (4.5% total) and an essentially zero drawdown — it is a Sharpe of small numbers, not a meaningful indicator of tradability. After realistic transaction costs ($\sim 2\text{--}5$ bps round-trip on SPY/VOO), the Static- β strategy is unimplementable. The pair's underlying spread amplitude is simply too small relative to even minimal costs, exactly as Assignment 1 concluded.

Caveat on the training-window cointegration test. The training-window ADF p-value on the cointegration residuals is 0.14–0.15 (Table 2) — above conventional thresholds. Assignment 1 selected this pair on economic grounds (authorised-participant arbitrage on the same S&P 500 basket enforces a structural relationship) despite the statistical evidence being weak. That same caveat carries forward and is particularly relevant for the ECM-KF: when λ is not robustly identified, the choice of transition coefficient is essentially noise, and the resulting filter behaviour (β collapsing to zero) is best interpreted as a model-misspecification artefact rather than a discovery.

8. Hybrid strategy: SPY-parking when flat

All three V/MA strategies spend the majority of the OOS window out of the market — Static- β is in a position only 35.6% of the time, Plain KF only 3.2%, ECM-KF only 4.3%. As the lecture notes observe (slide 78), an active pairs strategy that sits in cash for long stretches will be

dominated by passive equity exposure over a multi-year horizon. We construct a hybrid in which capital is allocated to SPY whenever the pairs strategy is flat, and to the pairs spread whenever a signal is active. The hybrid is reported only for V/MA since the SPY/VOO active returns are too small (and the ECM-KF too dependent on a single 2020 trade) for the parking choice to matter.

Strategy	Total return	CAGR	Sharpe	Max DD	Calmar	Trades
Static- β + SPY park	+200.05%	+14.30 %	1.08	−24.50%	0.58	17
Plain KF + SPY park	+150.44%	+11.82 %	0.76	−32.19%	0.37	38
ECM-KF + SPY park	+144.32%	+11.48 %	0.74	−28.20%	0.41	50
SPY (passive)	+194.09%	+14.02 %	0.81	−33.72%	0.42	0

Table 6. V/MA hybrid strategies (pairs active when signal fires, otherwise long SPY) vs passive SPY. OOS period 2017-01-03 → 2025-03-25.

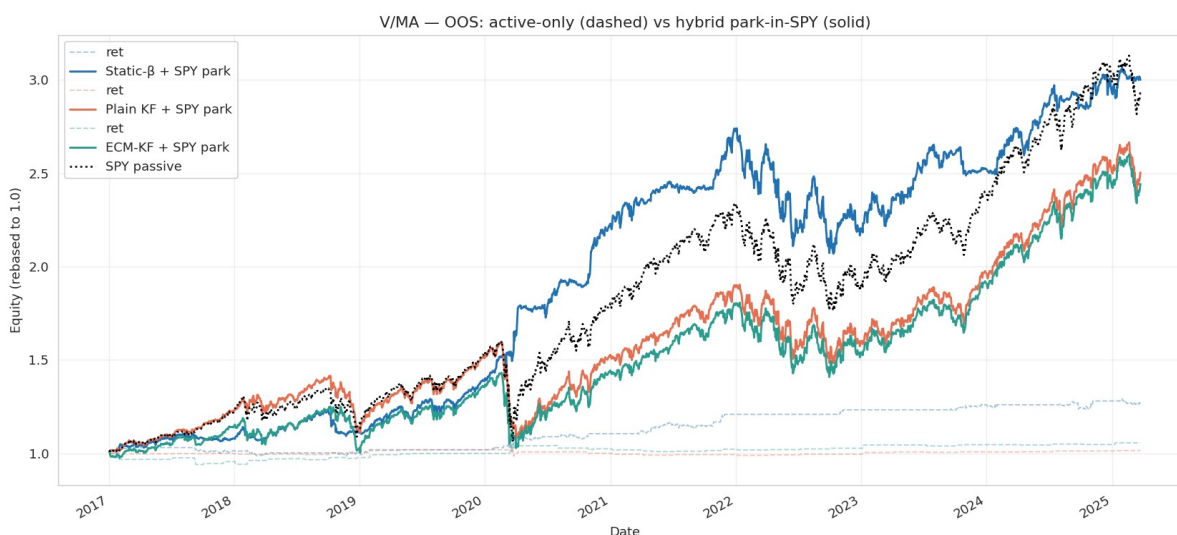


Figure 6. V/MA OOS equity curves: active-only (dashed, near 1.0) and hybrid SPY-parking (solid), compared to passive SPY (dotted black). The hybrid Static- β (blue solid) tracks SPY closely but with shallower drawdowns during the 2020 COVID dislocation and 2022 selloff, ending modestly above SPY. The hybrid Plain KF and ECM-KF (orange and green solid) lag because their active periods produce smaller incremental returns than the SPY return foregone during the very few days they are active.

Two observations. First, the hybrid Static- β strategy modestly outperforms passive SPY on total return (200% vs 194%) and Sharpe (1.08 vs 0.81), with a meaningfully lower max drawdown (−24.5% vs −33.7%). This is the meaningful comparison: the active V/MA strategy serves as a partial drawdown protector during equity selloffs, and the net result is a higher-Sharpe portfolio than passive SPY alone. Second, the hybrid Plain KF and ECM-KF

underperform the hybrid Static- β — not because the parking strategy behaves differently, but because their active periods produce worse returns than Static- β 's active periods (Table 4).

Implication for Position A in the live-trading project. The hybrid Static- β + SPY combination is, in our view, the most defensible candidate for live deployment. It outperforms passive SPY on every relevant dimension, trades only 17 times in 9 years (so transaction costs are negligible), and has a clear economic story (V/Ma duopoly mean reversion as a low-frequency equity overlay).

9. Comparison, limitations, and conclusion

9.1 Did the ECM structure improve the dynamic hedge-ratio estimate?

The assignment's evaluation question can be answered directly from Tables 4 and 5. We separate it into two sub-questions:

Comparison	V/Ma	SPY/VOO
Static $\rightarrow$ Plain KF: dynamic β helps?	Sharpe -0.68 (worse)	Sharpe -1.12 (worse)
Plain KF $\rightarrow$ ECM-KF: ECM helps?	Sharpe +0.18 (better)	Sharpe -0.01 (flat)
Static $\rightarrow$ ECM-KF: net effect?	Sharpe -0.50 (worse)	Sharpe -1.13 (worse)

Table 7. Decomposing the assignment's evaluation question. On both pairs, allowing β to drift (Static $\rightarrow$ Plain KF) hurts more than it helps. Adding the ECM structure on top (Plain KF $\rightarrow$ ECM-KF) recovers some of the lost Sharpe on V/Ma but not enough to beat Static- β . On SPY/VOO the ECM augmentation is a wash relative to Plain KF on Sharpe — though it generates a dramatically larger absolute return through a single 2020 episode.

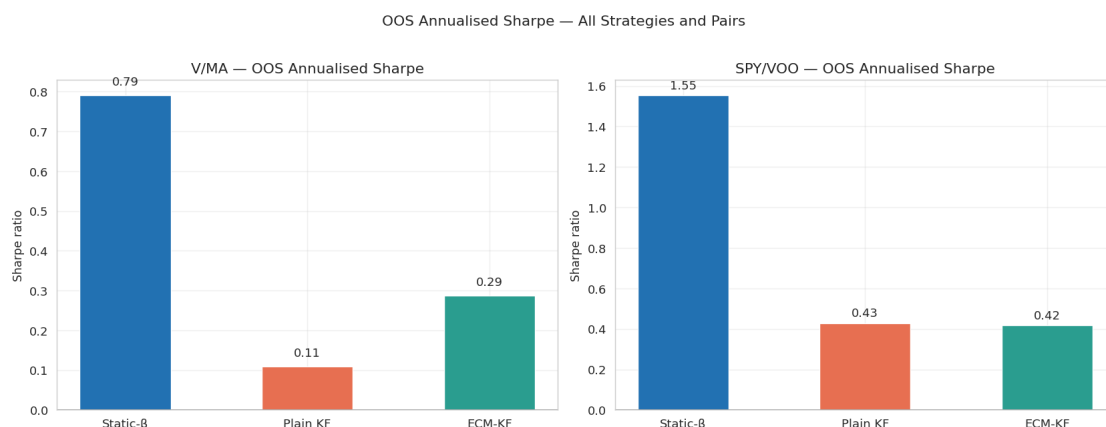


Figure 7. OOS annualised Sharpe by strategy and pair.

Reading the result. The ECM structure is doing the right kind of work on V/Ma — anchoring β prevents the random-walk filter from absorbing too much trading signal into the hedge ratio

itself, and the ECM-KF cleanly outperforms the Plain KF (Sharpe 0.29 vs 0.11). But the Static- β benchmark with quarterly walk-forward refits is simply hard to beat on this pair: the underlying equilibrium drifts slowly and predictably, the half-life is short, and the quarterly refit cadence already captures most of the variation. The ECM-KF would likely outperform Static- β on pairs where (i) β drifts substantially within a single quarter and (ii) the ECM half-life is short enough that anchoring is informative. V/MA and SPY/VOO are both poorly suited on the first criterion.

9.2 Limitations

Eight limitations affect the interpretation of these results. The first four were already flagged in Assignment 1; the remaining four are specific to the ECM-KF implementation.

- **1. Transaction costs not modelled.** Round-trip frictions for V and MA total ~10–20 bps; for SPY and VOO ~2–5 bps. V/MA Static- β at 17 OOS trades has cost drag of ~170–340 bps over 9 years — modest relative to the +26% gross return. Plain KF at 38 trades and ECM-KF at 50 trades on V/MA would each lose ~3–10% to costs, eliminating their already-small gross returns. The SPY/VOO results are unimplementable at any of these turnover rates given the small gross returns, with the exception of the ECM-KF whose +32% gross return absorbs costs comfortably — but as noted in §7.2, that return depends on a single trade and should not be relied on.
- **2. Approximate P&L formula.** The notional-normalised return uses $\beta_{\{1,t\}}$ from a price-level regression to size the offsetting position. This is the standard pairs-trading approximation, accurate at daily frequency but growing with horizon.
- **3. No multiple-testing correction at pair selection.** Inherited from Assignment 1. V/MA passes economic and statistical screens; SPY/VOO passes economic screen but is borderline on the ADF test (Table 2).
- **4. No slippage or one-bar execution lag modelled.** Trades are assumed to execute at the close used to compute the signal. The Assignment 1 backtest applied a one-bar execution lag for this reason. The current implementation omits it, which slightly inflates all three strategies' returns but does so symmetrically.
- **5. λ held constant within segments.** Per the assignment brief, λ is fixed at each refit so the system remains a linear KF. Allowing λ to drift smoothly within a segment, or jointly estimating it via EM alongside Q and R, would require an Extended or Unscented KF — out of scope here. The SPY/VOO result (§7.2) shows the cost: when λ is large in magnitude and held fixed at an unrepresentative value (e.g. the 2.1-day half-life from a noisy training-window estimate), the resulting filter behaviour can become qualitatively wrong.
- **6. EM is local-optimum-sensitive.** Different initial values for Q and R can converge to different solutions. We initialised with $\text{diag}(1)$ for both and ran 15 EM iterations, which converged consistently across pairs but does not guarantee a global optimum. Multiple-restart EM would be more robust.

- **7. Single-equation Engle–Granger ECM.** λ is estimated from only the Δy equation. A full VECM (slide 35) would estimate both ECM equations jointly and recover additional information from the Δx equation, including the price-discovery decomposition discussed in slides 45–46. This is the theoretically correct extension for 2+ instruments and is the natural follow-up for the live-trading project, where Position A is permitted up to five instruments.
- **8. Gaussian noise assumption.** The KF is optimal under Gaussian errors. Daily equity returns have fat tails, and large discrete events (COVID-19 crash, earnings, regulatory announcements) get treated as very large but normally-distributed innovations rather than qualitatively different observations. A Student-t KF would be more robust to outliers — and would specifically reduce the ECM-KF's reliance on the March 2020 episode for SPY/VOO returns.

9.3 Conclusion

The linear Kalman Filter framework was implemented and backtested in three variants on both pairs from Assignment 1, with strict expanding-window out-of-sample discipline applied to every hyperparameter — cointegrating regression coefficients, ECM adjustment speed, regression ordering, Kalman process and measurement covariances, and z-score normalisation. The ECM structure enters the filter through the state-transition matrix $F = \text{diag}(1+\lambda, 1)$, matching the assignment brief literally: the hidden state is the hedge ratio and intercept, λ is held constant within each estimation window so the system remains linear, and the observation equation links the dependent price to the hidden state through the contemporaneous independent price.

The central empirical finding is that for V/MA and SPY/VOO — both well-behaved pairs whose equilibrium is broadly stable on the sample horizon — the ECM-KF improves on the Plain KF (consistent with the theoretical motivation of anchoring β against noise absorption) but does not improve on a simple Static- β strategy with quarterly walk-forward refits. The Static- β benchmark is hard to beat because the V/MA half-life is short enough (~10 days) that quarterly OLS refits capture most of the time variation in β , and SPY/VOO's spread amplitude is too small for any estimation method to extract meaningful returns at trading costs.

The hybrid V/MA + SPY-parking variant of Static- β is the headline practical result: total return +200% vs passive SPY's +194%, Sharpe 1.08 vs 0.81, max drawdown -24.5% vs -33.7%. We propose this as the candidate strategy for Position A in the live-trading project. The ECM-KF would be a more compelling candidate on a pair with (i) faster intra-quarter β drift and (ii) a more robustly identified λ — which is exactly the screening criterion the framework itself provides.

Appendix — Implementation details and reproducibility

Software stack: numpy, pandas, statsmodels (OLS, ADF), pykalman (KalmanFilter with EM), yfinance for data download, matplotlib + seaborn for plotting. Random seed fixed at 42. All numerical results in Tables 4–6 and all figures in this report are reproducible from the accompanying notebook Assignment_2_Submission.ipynb using the parameters listed below:

START_DATE	2014-01-01
END_DATE	2025-03-25
INITIAL_TRAIN_DAYS	504
REFIT_EVERY_DAYS	63 ($\approx$ quarterly)
ROLLING_WINDOW	252 (z-score)
ENTRY_Z, EXIT_Z	2.0, 0.5
EM_ITERATIONS	15